

MSCR Stereo Camera Calibration Report

Date: 2026-06-03T13:04:17 Sensor: RealSense D435i IR stereo Mount: fixed

Parameter	Value
Resolution	1280 x 720 px
Stereo RMS reprojection	1.2442 px (ACCEPTABLE)
Mean epipolar error	0.2925 px (PASS)
Max epipolar error	2.1745 px
Image pairs used	44
Left fx / fy	632.324 / 632.536 px
Right fx / fy	633.704 / 633.213 px
Stereo baseline	49.520 mm
Inter-camera rotation	0.5888 deg
Horizontal / Vertical FOV	90.69 / 59.29 deg

Quality: **ACCEPTABLE** (RMS = 1.2442 px) Epipolar: **PASS** (mean = 0.2925 px)

Thresholds: RMS below 0.5 px (excellent), epipolar below 1.0 px (pass). Results above 1.5 px / 2.0 px respectively indicate insufficient pose diversity or board motion during capture.

1. Intrinsic Parameters

1.1 Left IR Camera

$f_x = 632.3241$ px $f_y = 632.5361$ px $c_x = 647.8026$ px $c_y = 369.6477$ px FOV = 90.69 x 59.29 deg

	col 0 (x)	col 1	col 2 (y)
row 0	632.324113	0.000000	647.802555
row 1	0.000000	632.536133	369.647668
row 2	0.000000	0.000000	1.000000

k1	k2	p1	p2	k3
-0.05151750	0.10123990	0.00495700	0.00166673	-0.10594484

Brown-Conrady model. Radial: k1, k2, k3. Tangential: p1, p2.

1.2 Right IR Camera

$f_x = 633.7036$ px $f_y = 633.2131$ px $c_x = 652.0696$ px $c_y = 364.3636$ px FOV = 90.57 x 59.24 deg

	col 0 (x)	col 1	col 2 (y)
row 0	633.703638	0.000000	652.069555
row 1	0.000000	633.213057	364.363601
row 2	0.000000	0.000000	1.000000

k1	k2	p1	p2	k3
-0.04947404	0.05566914	0.00465741	0.00160330	-0.01351836

Brown-Conrady model. Radial: k1, k2, k3. Tangential: p1, p2.

1.3 Factory Firmware vs This Calibration

Factory intrinsics shown for reference only. The values above should be used in the control pipeline.

	Cal Left	FW Left	Delta	Cal Right	FW Right	Delta
f_x	632.324	639.508	-7.184	633.704	639.508	-5.804
f_y	632.536	639.508	-6.972	633.213	639.508	-6.295
c_x	647.803	642.885	+4.918	652.070	642.885	+9.185
c_y	369.648	355.650	+13.997	364.364	355.650	+8.713

2. Stereo Extrinsics

Rotation R (left to right IR) and translation T:

R[row,col]	col 0	col 1	col 2
row 0	0.99997341	-0.00079723	-0.00724874
row 1	0.00084969	0.99997346	0.00723638
row 2	0.00724278	-0.00724235	0.99994754

T = [-49.51949, 0.26465, -0.05489] mm |T| = 49.5202 mm Rotation = 0.5888 deg

Rotation angle below 0.5 deg indicates parallel-mounted cameras (nominal for the D435i). Values above 1 deg may indicate a mounting issue.

3. Reprojection Error Analysis

Per-pair RMS for left (blue) and right (red) after iterative outlier rejection. Orange bars exceed the 1.0 px limit. Right panels show per-corner error vectors with 1-sigma and 2-sigma ellipses. A symmetric, tight cluster near the origin indicates well-conditioned calibration.

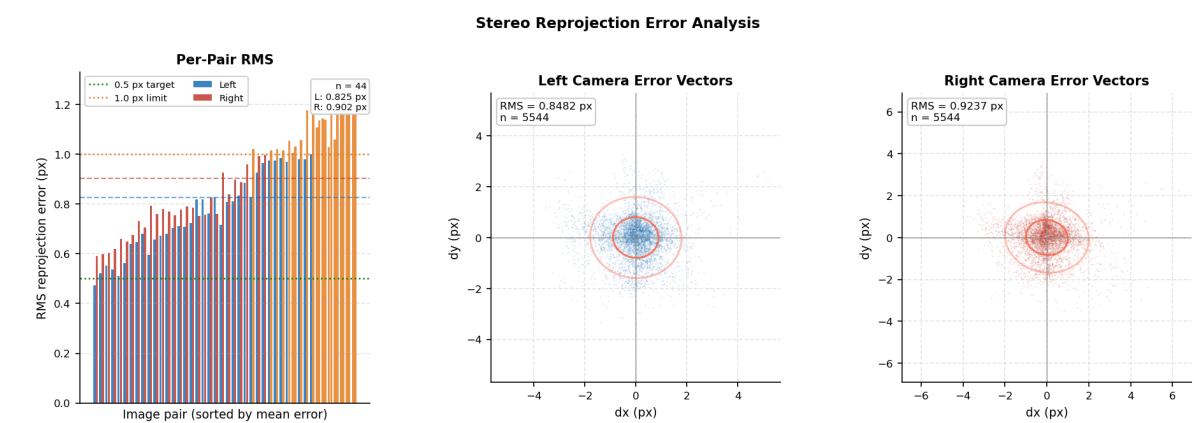


Figure 1 - Per-pair RMS (left) and per-corner error vectors for left and right cameras.

4. Stereo Rectification Verification

Original IR pair (top) and rectified pair with epipolar lines (bottom). After rectification, any corresponding scene point must lie on the same horizontal scanline in both views. Vertical offset is residual calibration error.

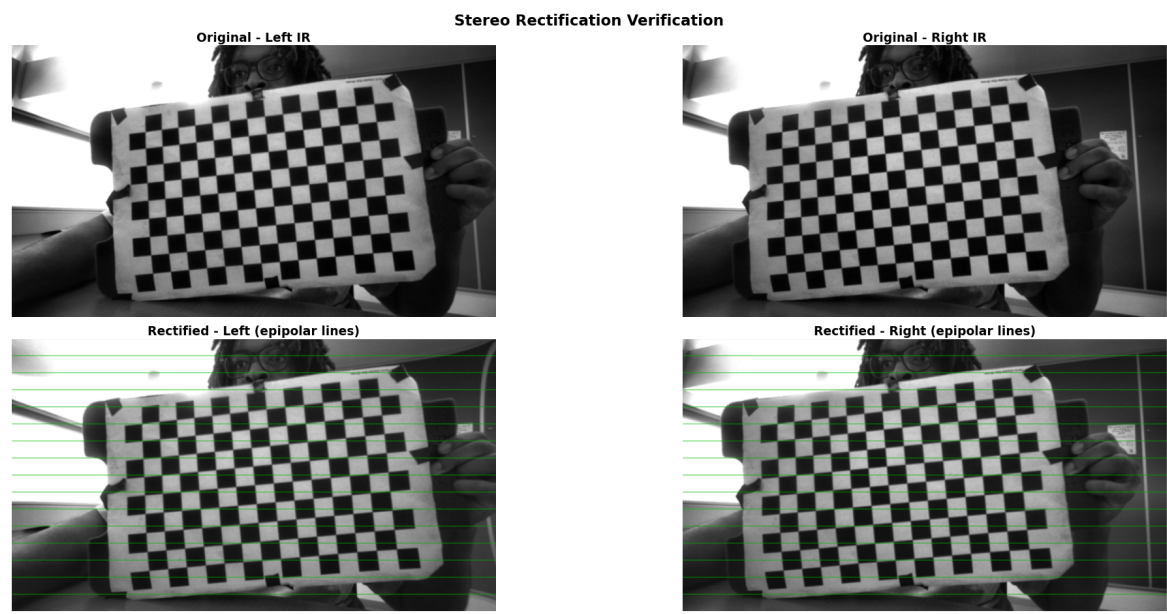


Figure 2 - Original and rectified IR pair. Green lines are epipolar lines.

5. Lens Distortion Maps

Per-pixel undistortion displacement for left and right IR sensors. Arrows show the direction and magnitude of the correction at a sampled grid. Barrel distortion dominates at the sensor periphery as expected.

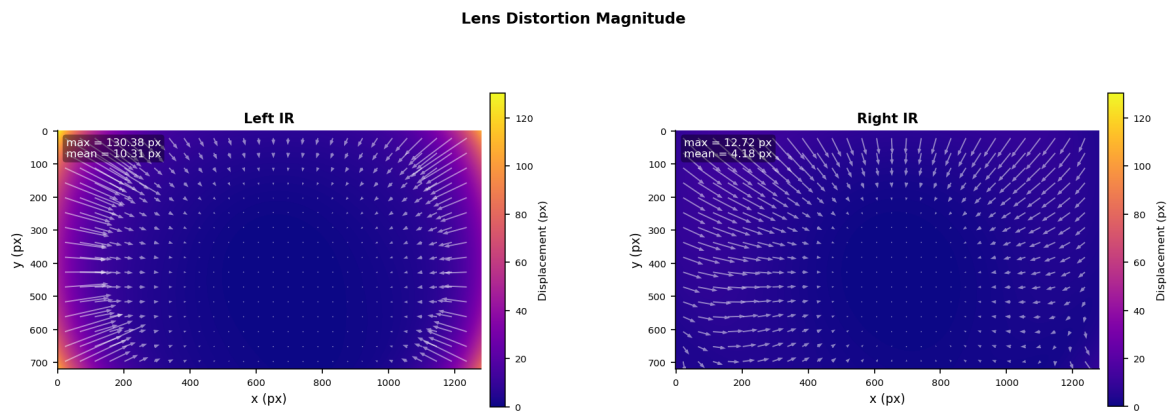


Figure 3 - Distortion displacement magnitude (px) for left and right cameras.

6. Calibration Pose Coverage

Estimated checkerboard positions in the left camera frame across all accepted image pairs. Broad spatial coverage across X, Y, and Z produces a well-conditioned calibration; clustering in one region can inflate error at other positions.

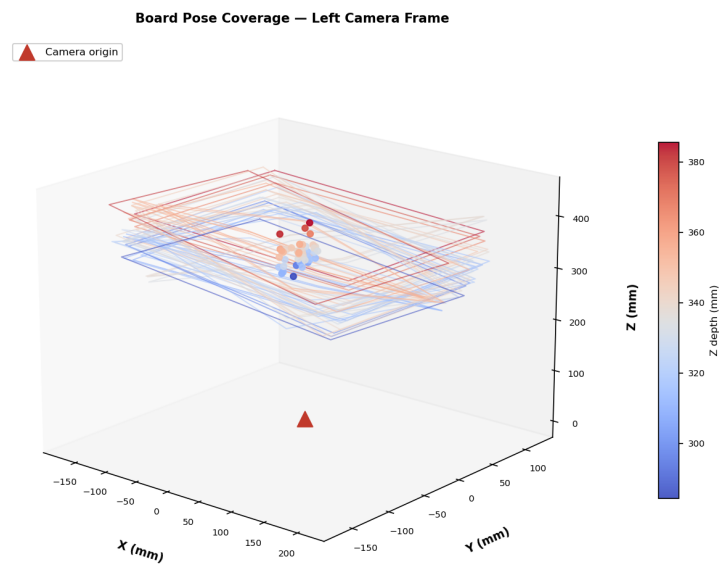


Figure 4 - Board pose distribution in the left camera frame (mm).

7. Depth Resolution for MSCR Control

Depth $Z = f_x * B / d$, where $B = 49.52$ mm and $f_x = 632.3$ px. At 200 mm working distance, depth resolution is 1.277 mm per disparity pixel; sub-pixel disparity estimation is required for sub-millimetre endpoint tracking at this range.

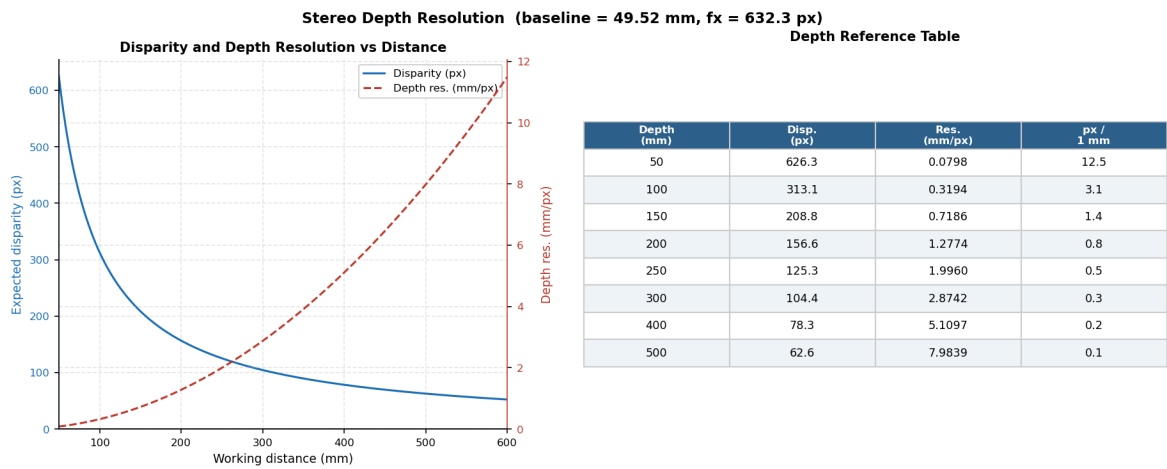


Figure 5 - Disparity and depth resolution vs working distance, with reference table.